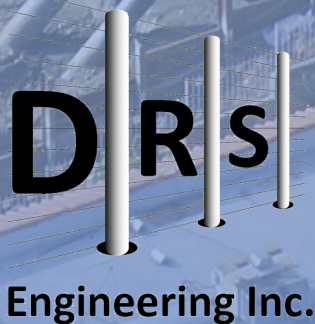


STATEMENT OF QUALIFICATIONS

DRS ENGINEERING INC.

EARTH RETENTION AND FOUNDATION DESIGNERS





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ABOUT DRS ENGINEERING

After working in earth retention construction in the Southern California market for almost a decade, founding principal, Dr. David Salter, recognized the need for highly competent, efficient and responsive earth retention design services that minimized design time without sacrificing quality. Born from the desire to improve deliverable consistency and quality, Dr. Salter established DRS Engineering in 2009 to meet the needs of owners, developers, general and specialty contractors with challenging geo-structural issues related to excavation support, slope stabilization and specialty foundations.



DRS Engineering Inc. is a specialist geo-structural engineering consultancy providing services related to earth support systems and specialty foundations. Our design services fall in three broad categories: Temporary Shoring, Permanent Earth Retention and Specialty Foundations. Founded on over 40 years of experience we



provide our clients with the most economical solutions based upon innovative approaches created to meet the project challenges. Our clients range from individual homeowners to multi-national developers, and we provide full project support from concept through detailed design, plan and specification preparation, bid processing and construction supervision.

With engineering licenses in California, Washington and Oregon, DRS Engineering is the premier west coast earth retention designer utilizing state of the art engineering techniques to provide the most efficient and technically competent services in the region.

Our expertise and efficiency is born out of many years of experience in the construction of earth retention systems that enables us to focus on the critical issues for contractors and owners. We provide efficient practical solutions in a timely manner. Our practical knowledge of what is critical and what is not enables us to safely respond to changes in conditions without delays caused by the need to resort to calculator or computer.



In the words of Karl Terzaghi *“Theoretical reasoning alone does not suffice to solve the problems which we are called upon to tackle”* which is why our experience is critical to ensuring the most-cost effective solutions for your earth retention challenge.

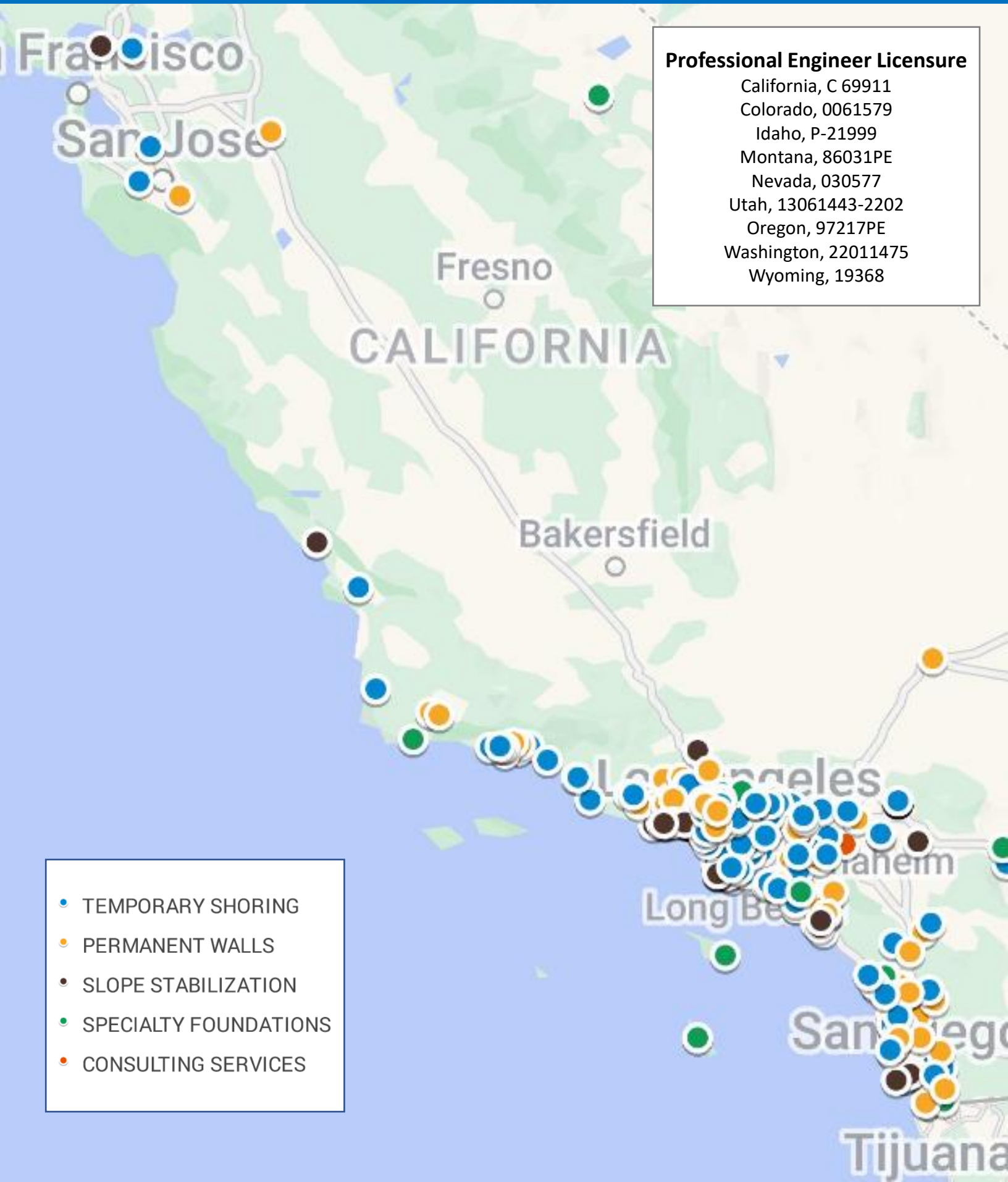
Our goal is to provide timely **“SUPPORT YOU CAN RELY ON”** within budget and to earn your trust for repeat business in the future.

Core Values

- Technical competence
- Responsiveness to clients needs
- Timely deliverables
- Collaborative spirit
- Respectful interactions
- Innovation

OUR PROJECTS

OVER 500 PROJECTS SINCE 2009



PARTNERSHIPS



Vitruvian Design Group
Access Limited Construction
Alliance Residential
Apex Rockfall
ARC City
Arxis Design Studio
Austin Family Development, LLC
Awbrey Cook McGill
C W Howe, Inc.
Calex Engineering Co.
Center Inter'l Investments
Condon-Johnson & Associates
Cornerstone Company
Crest Real Estate, LLC
CRP Affordable Housing
D.J. Scheffler, Inc.



D.R. Horton
DAIGC
Dakota Drilling
Diamond West Incorporated
Excel Engineering
Garnett DePasquale Projects
Geostablization International
Geobruigg
GroundForce
Groundtech
Guardian Paradise, LLC
Harridge Development Group
InsitePG
KPFF



Malcolm Drilling Company, Inc.
Jamison Properties, LP
JD Group (Los Angeles Office)
KGC Drilling and Shoring
Kindness General Contractors, Inc.
La Peer Hotel Owners, LLC
MCRT California Construction LP
Mill Creek Residential Trust (MCRT)
MNS Engineer, Inc.
Norris Construction
Onni Contracting
OP Real Estate Development Group
Praxis Engineering
R. C. & Associates
RCI Builders
Swinerton L.A.



TagFront
United Building Company
The J. Paul Getty Trust
Westcord Commercial Real Estate





TEMPORARY SHORING

SOLDIER PILE AND LAGGING

SOIL NAILING

SHEET PILES

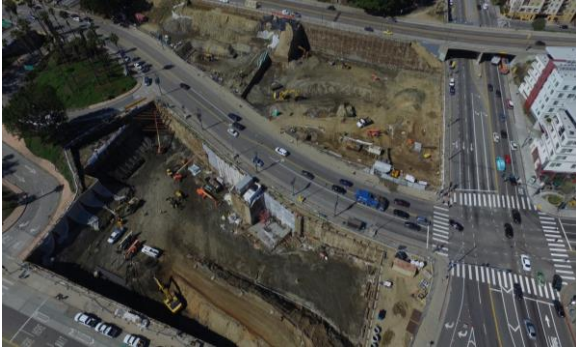
DEEP SOIL MIXED

SECANT PILE



TEMPORARY SHORING

DRS Engineering has extensive experience in the design of all types of temporary shoring systems ranging from simple shallow cantilever soldier pile and lagging shoring to complex deep excavation shoring using rakers, bracing, temporary tiebacks or soil nails. Our experience enables us to quickly determine the optimum shoring system for your project given your unique ground conditions and physical site characteristics. We provide you with designs that will ensure a safe and efficient excavation process with minimum risk to surrounding properties.



Project: La Plaza Y Cultura, Los Angeles CA

Work Type: Temporary Soldier Pile and Lagging Shoring

Client: Condon Johnson

Description: This 75,000SF shoring project, located in the heart of downtown Los Angeles, required up to 55ft deep excavation adjacent to major city streets and freeway on-ramps. In addition, the site sat above two major metro tunnels with critical movement criteria.



Project: Burbank Reservoir, Burbank CA

Work Type: Temporary Soil Nailing

Client: Avar

Description: This 12,000SF soil nailing system retained up to 35ft of variable ground including quartz bedrock, alluvial soils and artificial fills in order to facilitate the construction of a new water reservoir for the City of Burbank



Project: Tramonto Villas, Pacific Palisades CA

Work Type: Temporary Tieback and Raker Braced Shoring

Client: ETCO Homes

Description: Located on the site of a recent major landslide, this residential development involved up to 60ft deep excavation adjacent to the upslope property lines and required significant raker braced shoring to support high-rise structures above the excavation as well as internal temporary tieback shoring.



Project: Modera Hollywood

Work Type: Temporary Tieback and Raker Braced Shoring

Client: Millcreek Residential

Description: Located in the heart of Hollywood, this residential development involved up to 55ft deep excavation adjacent to historic building and busy city streets. Over 250 high capacity temporary tiebacks were used to support this major excavation.

A high-angle photograph of a construction site on a hillside. In the foreground, several large, white, cylindrical concrete piles are visible, some with yellow and black striped sections. A worker in a bright yellow safety vest is standing on a platform near one of the piles. In the background, there are trees, a road, and a body of water under a clear blue sky. The text "PERMANENT RETAINING WALLS" is overlaid in large white letters.

PERMANENT RETAINING WALLS

PILE WALLS
SOIL NAIL WALLS
STEM WALLS



PERMANENT RETAINING WALLS

DRS Engineering provides designs for a wide range of permanent retaining wall systems ranging from simple low CIP stem walls to complex tall ground anchor supported or soil nail supported walls. The choice of the most suitable retaining wall system can have significant impact on construction costs and our experience enables us to quickly determine the best solution for your project.



Project: Enchanted Way – Pacific Palisades CA

Work Type: Permanent Anchor Supported Soldier Pile Wall

Client: Homeowners

Description: After a significant landslide carried away the rear yards of these valuable properties in the Pacific Palisades area of Los Angeles, a permanent soldier pile wall supported by permanent ground anchors was installed to safeguard the structures and enable the rear yards to be re-created.



Project: Cambria Hotel, Calabasas CA

Work Type: Permanent Soil Nail Walls

Client: GroundForce Inc.

Description: Excavation into the toe of the Santa Monica Mountains was needed for the creation of the new Cambria Hotel in Calabasas. The Topanga and Modelo formations were suited to soil nailing and the sculpted and stained 12,000SF wall now forms the backdrop to the hotel pool and courtyard.



Project: 801 Nimes Place, Bel Air, CA

Work Type: Permanent Caisson Fill Wall

Client: Beverly Hills Investments LLC

Description: The creation of a new elevated building pad on the side of an existing hill required the placement of up to 50ft of fill. DRS Engineered a caisson wall supported by deadmen anchors constructed at various elevations as backfill proceeded.



Project: Microvention Building, Aliso Viejo, CA

Work Type: Permanent Anchor Supported Soldier Pile Wall

Client: NMG Geotechnical

Description: Grade separation between the proposed office building and adjacent parking garage required a 45ft high permanent retaining wall. The wall was created using soldier beams supported by two tiers of permanent ground anchors. Wood lagging provided soil support during the excavation and the final permanent 12" shotcrete facing was placed monolithically, attached to the soldier piles, with concrete headed studs.

An aerial photograph of a modern, multi-level building complex built into a hillside. The building features white walls, large glass windows, and several swimming pools. The surrounding landscape is lush with greenery, and the background shows a residential area on a hillside under a clear blue sky. The text is overlaid on the image in white, bold, sans-serif font.

SLOPE STABILIZATION

CAISSONS & SHEAR PINS
PERMANENT GROUND ANCHORS
SOIL NAILING
DRAPERY SYSTEMS
ROCKFALL BARRIERS
DEBRIS FLOW CONTROL



SLOPE STABILIZATION

Slope failures and slope stability improvement are significant issues in Southern California as development is pushed into more marginal, yet aesthetically desirable terrain. DRS offers a wide range of design solutions for slope stabilization issues ranging from simple regrading and wall systems to complex caisson, micropile or ground anchor systems for the global stabilization of large slopes. DRS also specializes in the design of systems for the prevention and control of natural hazards such as rockfalls and mudflows using specialized netting and barrier systems.



Project: South Shore Church Dana Point CA

Work Type: Permanent Ground Anchors

Client: LGC Geotechnical Inc.

Description: Deep seated instability in the slopes below the proposed church expansion required the use of 35 3ft diameter caissons, up to 75ft deep, and 216 permanent ground anchors up to 140ft in length, each with a design load of 310kip. Ground anchors pre-stressed the earth through a three tier grade beam system supported on the caissons.



Project: 944 Airole Drive, Bel Air CA

Work Type: Tecco and Soil Nails

Client: Crestlloyd LLC

Description: To create “**THE ONE**” (as the most expensive house in the US has been called) the top of a prominent hill in the Bel Air neighborhood of LA was removed and the associated slopes stabilized with a Tecco mesh and soil nail system. The open structure of the mesh system enabled planting on the stabilized slopes giving a natural look to the pedestal supporting the magnificent structure



Project: Arco #9616 Topanga, CA

Work Type: Shear Pins, Tecco and Ground Anchors

Client: Fiedler Group

Description: The abandoned gas station on the scenic PCH in Topanga CA has been an eyesore for many years. The old cut slopes above the site were deteriorating and unstable. Through a combination of permanent ground anchors, concentric steel pipe shear pin piles and Tecco and soil nails, the slope was able to be stabilized and a new gas station opened.



Project: Palos Verdes Water Reliability Project

Work Type: Rockfall Barriers

Client: Access Limited Construction

Description: Steep rocky slopes above the proposed new water distribution facility on the Palos Verdes Peninsula posed a threat to the new facility. Several Geobrugg GBE 500A-R rockfall barriers were constructed at the base of the slope. The barriers were 6ft high based on a computer simulation of potential rock fall trajectories at the site.

SPECIALTY FOUNDATIONS

CAISSON FOUNDATIONS
MICROPILE FOUNDATIONS
DISPLACEMENT PILES



SPECIALTY FOUNDATIONS

Specialty foundations are often required for new construction and or existing construction retrofit where conventional concrete foundations are not appropriate. DRS Engineering provides design services for a range of pile foundation systems including; conventional caissons, micropiles and displacement piles.



Project: Coachella Micropiles

Work Type: Micropile foundations

Client: Magco Drilling

Description: Micropiles were used for compression and tension resistance to overturning forces on 72 transmission towers thru the Coachella valley in California. Micropiles were a combination of open hole or self drilled piles based on local ground conditions at each tower



Project: Entrada

Work Type: Caisson Foundations

Client: D J Scheffler

Description: The development of a major new hotel and retail complex on the waterfront in Santa Barbara required the use of deep foundations to mitigate liquefaction in the sandy water bearing soils. D J Scheffler installed early verification test piles using an augercast technique



Project: Grand Canyon Skywalk

Work Type: Micropile Foundations

Client: D J Scheffler

Description: Suspended 3600ft above the Colorado river, the Grand Canyon Skywalk in the Hualapai Reservation is a Incredible attraction. Before the tourists could safely walk out on this cantilevered glass structure, the micropile foundations anchoring the structure to the solid rock were designed by DRS and installed and tested by DJ Scheffler.



An aerial photograph showing a large, light-colored, eroded area on a hillside, likely a landslide or a quarry. The surrounding area is covered in dense green vegetation. At the bottom of the image, a residential neighborhood with various houses and buildings is visible. The text is overlaid on the center of the image.

CONSULTING SERVICES

EXPERT WITNESS
GEOTECHNICAL NUMERICAL MODELING ANALYSIS



CONSULTING SERVICES

Expert Witness

Based on many years of practical experience in designing and constructing slope stabilization systems, shoring, retaining walls and specialty foundation systems, DRS Engineering is able to provide clarity and insight into the key issues that are often at the heart of litigation in these areas.



Project: Tramonto Villas, Pacific Palisades CA

Parties: California Landmark Group v Jerome Nash

Retaining Attorney: Liner, LLP Los Angeles, CA

Deposed: Yes

Case Summary: Provided expert witness testimony on the requirement for and design of a permanent earth retention system to protect existing apartment complex from the Revello landslide in the Pacific Palisades. Case Settled prior to trial.



Project: Icon, San Diego, CA.

Parties: Ledcor v Tower Glass

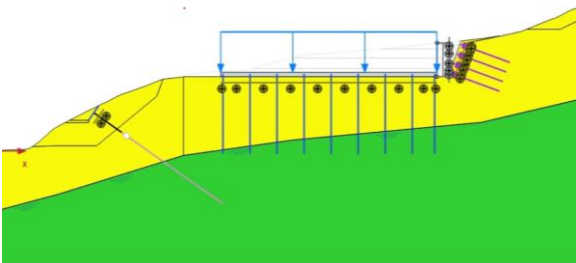
Retaining Attorney: Procopio, San Diego CA

Deposed: Yes

Case Summary: Provided expert witness testimony during arbitration regarding the causes of a collapse of a temporary soil nail wall system at the Icon project in San Diego where fiberglass soil nails were utilized.

Numerical Modeling Analysis

DRS provides a fully integrated numerical modelling analysis developed from over 7 years of in-house experience. We have routinely delivered and analyzed complex 2D and 3D models for fast-paced projects including sites with difficult ground conditions.



Project: Vista Minor

Work Type: Slope Stabilization

Client: Geostabilization International

Description: Up to 50 feet of landslide deposits overlie the highly sheared marine sedimentary bedrock at this exclusive homesite development in Big Sky Montana. DRS performed full-scale 2D static and dynamic analyses for the stabilization system proposed to mitigate against landslide hazards for the new residence. The 2D analysis allowed for a more efficient final design that reduced the costs up to 50% of the engineer's preliminary estimates.

OUR TEAM



OUR TEAM



David R. Salter
BSc PhD PE
Principle

Dr. Dave Salter has over 40 years of experience in the design and construction of earth retention systems and specialty foundations in the USA and Europe. Dr. Salter worked with the Keller Group in the UK before moving to the USA and joining Schnabel Foundation Company in 1993. As Project Manager in Washington DC and Area Manager in Los Angeles, Dr. Salter was responsible for the design and construction of a wide variety of earth retention projects. From 2003 to 2009, Dr. Salter managed the Los Angeles office of TerraCosta Consulting Group and was the Engineer of Record for multiple projects involving temporary shoring, permanent earth retention, landslide stabilization and specialty foundations. Since 2009, as owner and Principal Engineer of DRS Engineering Inc., Dr. Salter provides specialty consultancy services to Contractors, Consultants and Owners. Dr. Salter is a regular speaker at industry seminars and provides expert witness testimony on issues related to earth retention and foundation systems.



Jody Robinson
MS PE
Senior Project Engineer

Jody Robinson earned his bachelor's degree in civil and environmental engineering, and a master's degree in civil engineering with a focus in geotechnical and geostructural engineering from Mississippi State University. He began his career as a Senior Staff Geotechnical Engineer for Hart Crowser in Seattle, Washington, and most recently worked as a Regional Engineer for a geohazard specialty contractor. Jody has 7 years of geotechnical and geostructural engineering experience in temporary and permanent slope stabilization, deep excavation shoring projects, and landslide remediation projects. He has been an experienced practitioner in numerical modeling with the finite element method for the last 5 years. Jody's experience includes design and project management for projects with difficult site conditions such as; thick profiles of soft, compressible materials, fissured heavily over consolidated clays, talus, severe liquefaction, and lateral spread hazards.



Neil Ferguson
Project Engineer

Neil is a geotechnical engineer with a master's degree in geological engineering and a bachelor's degree in mechanical engineering. He has 6 years of professional experience in various aspects of geotechnical engineering, including slope stabilization, shoring, foundations, pavement, ground improvement, and numerical modeling. He has worked as a project engineer for GeoStabilization International, a design/build geotechnical contractor, and for GeoResources, LLC, a geotechnical consulting firm. He has also served as an engineer officer in the United States Army, leading teams of engineers and equipment operators in construction and military engineering projects. Neil currently resides near Seattle, Washington.

OUR TEAM



Olivier Dore
Designer

Olivier Dore has over twenty years of experience as a designer and project manager for structural projects and is highly skilled in the use of AutoCad for the production of technical civil, architectural and structural drawings. Olivier is responsible for the creation of DRS Engineering's well organized and accurate earth retention, and foundation plans and for implementing drawing standards. He is also responsible for project management for specific projects in the LA area. Olivier's experience with other professional disciplines including architects and structural engineers, enables him to provide the optimum presentation to convey complex information between disciplines. Olivier manages DRS Engineering's CAD database and digital drawing portfolios as well as the company's internal computer network system to ensure efficient data integrity and accessibility.



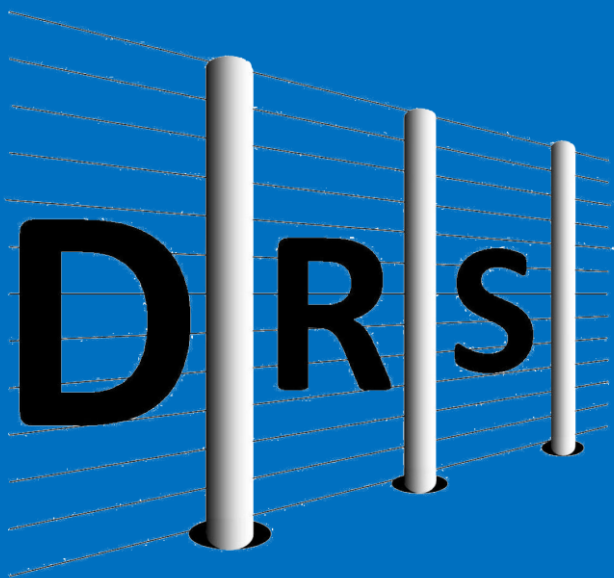
Sydney Kanazawa
Designer

Sydney Kanazawa is a designer for DRS Engineering with over five years of experience with AutoCAD. She is responsible for drawing production and helping to maintain the CAD library and standards. Sydney holds a B.S. in Civil Engineering with a focus on geotechnical and structural engineering from California Polytechnic State University, San Luis Obispo. Prior to joining DRS, she worked as a structural engineer in the Bay Area with experience in commercial, education, and residential projects..



Luke Bennett
MBA
Business Manager

Luke Bennett has over a decade of experience in finance and management, with a background in construction and real estate. Luke oversees DRS Engineering's operations with responsibilities including financial reporting, general ledger management, invoicing, banking and special projects. Prior to joining DRS, Luke worked in Commercial Real Estate Banking at Bank of America, serving on a team that held \$4.3B in loan balances in 2021. While at Bank of America, Luke also managed a team of 12, who contributed to the 3,000 PPP4 loans totaling more than \$1.5B in balances. Luke holds an MBA in Finance from Pepperdine University and a BA in International Studies from American University. Luke is a US Coast Guard veteran, and serves on the board of Mission Rowing.



Engineering Inc.

OFFICES IN
SANTA MONICA | SANTA BARBARA | SEATTLE